26900531 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSESTA302 - STATISTICAL COMPUTING USING PYTHON

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. Show the syntax for using an if-elif-else statement in Python with an example. [A] / [CO1]
2. State two advantages of using an IDE for Python programming. [U] / [CO1]
3. Differentiate between syntax error and logical error. [An] / [CO1]
4. How will you remove rows containing missing values, from a dataframe? [U] / [CO2]
5. Write the Python code to perform inner join of two datasets. [U] / [CO3]
6. Write the NumPy statement to create an array of five zeros. [A] / [CO2]
7. What is the main advantage of using relplot over scatterplot? [U] / [CO2]
8. What is the purpose of using swarmplot over boxplot? [U] / [CO2]
9. What does an approximately straight-line pattern in a Q–Q plot indicate? [U] / [CO3]
10. State the Inverse transform method. [U] / [CO4]

[2x7 = 14]

Part B

Short essay type questions

Answer any **4** questions

Each question carries **6** marks

11. Illustrate the basic data types in Python with suitable examples. [U] / [CO1]

12. With the help of suitable examples, explain about built-in functions in Python. [U] / [CO1]

13. Create a long-format sales DataFrame using dummy values, containing Region, Quarter and Sales, with duplicate Region-Quarter combinations. Use pivot_table to obtain total sales, melt the result back to long format, and draw a Matplotlib bar chart of total sales by region with title and axes labels. [A] / [CO2]

14. Create a DataFrame from a list of tuples: [('A', 25), ('B', 30), ('C', 35)] with column names ['Name', 'Age']. Print its shape, columns, and data types. [A] / [CO2]

15. Load the Seaborn titanic dataset and create a boxen plot of age across different class categories, using sex as hue. Write the python code used and interpret the result. [A] / [CO2]

16. Describe the layout and computational procedure of a one-way ANOVA. [U] / [CO4]

[6x4 = 24]

Part C

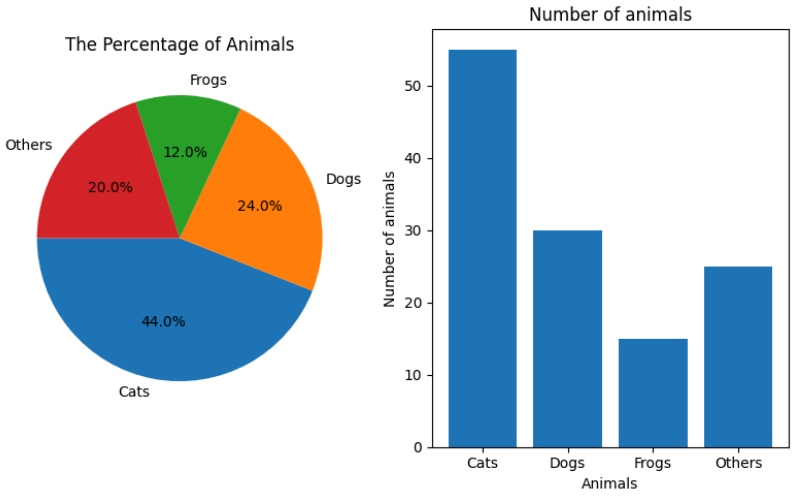
Essay type questions

Answer any **1** questions

Each question carries **12** marks

17. Write a python programme to display the given data in a diagram of the following form. [A] / [CO3]

Type of Animals	Cat	Dog	Frog	Others
Number of Animals	55	30	15	20



18. A psychologist measures the reaction time (in milliseconds) of 6 subjects under three different conditions (Condition 1, 2, 3) in a repeated-measures design: [A] / [CO4]

Subject	Condition 1	Condition 2	Condition 3
1	210	225	240
2	205	215	230
3	220	230	245
4	198	210	220
5	215	222	238
6	208	218	233

(a) Explain why sphericity needs to be tested before conducting a repeated-measures ANOVA on this data.
(b) Perform Mauchly's test of sphericity on this dataset.
(c) Conduct the repeated-measures ANOVA to test whether reaction time differs significantly across the three conditions. Write the complete Python code and interpret the results.

[12x1 = 12]